SISTEMAS OPERATIVOS

Aula Teórica 1

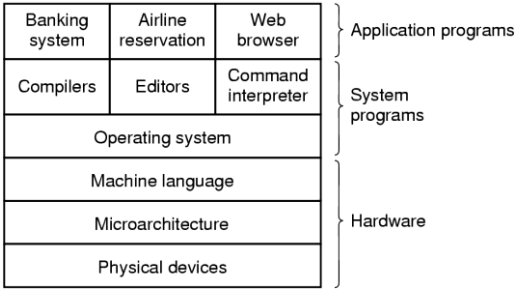
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1. What is an operating system
2. History of operating systems
3. The operating system zoo
4. Computer hardware review
5. Operating system concepts
6. System calls
7. Operating system structure

Introduction



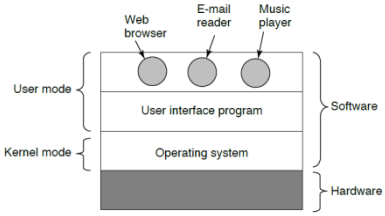


Figure 1-1. Where the operating system fits in.

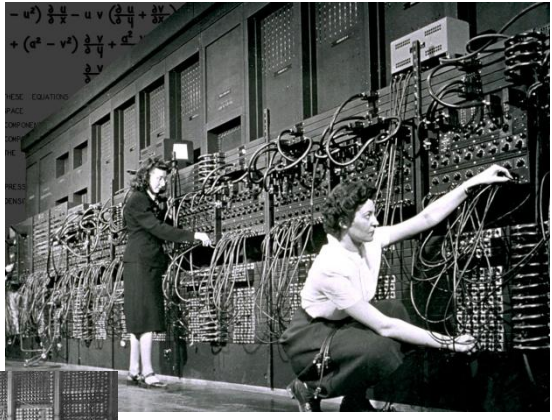
- A computer system consists of
  - hardware
  - system programs
  - application programs

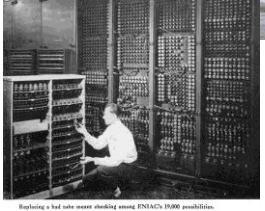
What is an Operating System

- It is an extended machine
  - Hides the messy details which must be performed
  - Presents user with a virtual machine, easier to use
- It is a resource manager
  - Each program gets time with the resource
  - Each program gets space on the resource

History of Operating Systems (1)

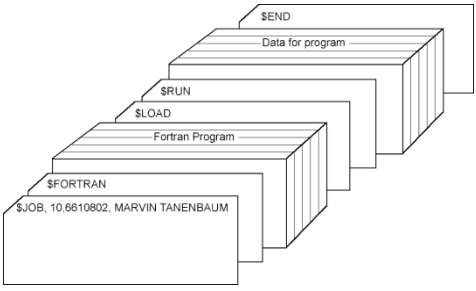
- First generation 1945 – 1955
  - vacuum tubes, plug boards
- Second generation 1955 - 1965
  - transistors, batch systems
- Third generation 1965 – 1980
  - ICs and multiprogramming
- Fourth generation 1980 – present
  - personal computers





Replacing a bad tube meant checking among ENIAC's 19,000 possibilities.

History of Operating Systems (2)



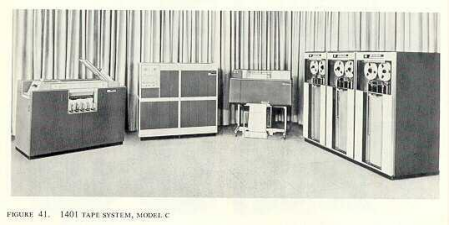
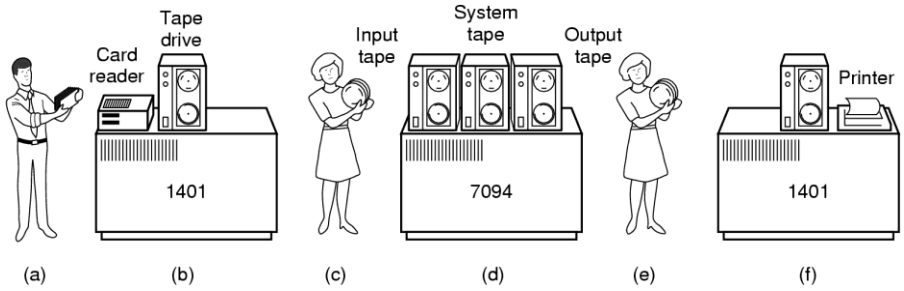


FIGURE 41. 1401 TAPE SYSTEM, MODEL C

- Structure of a typical FMS job – 2nd generation

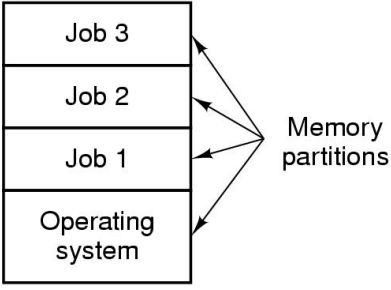
History of Operating Systems (3)



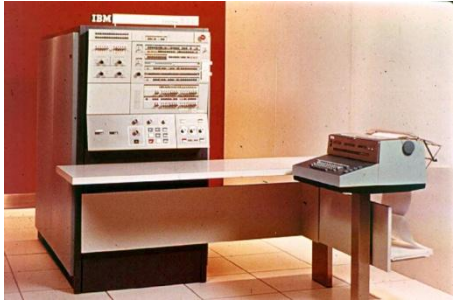
Early batch system

a) Programmers bring cards to 1401
b) 1401 reads batch of jobs onto tape
c) Operator carries input tape to 7094
d) 7094 does computing
e) Operator carries output tape to 7094
f) 1401 prints output

History of Operating Systems (4)







- Multiprogramming system
  - three jobs in memory – 3rd generation

History of Operating Systems (5)









Single individual can have is own personal computer– 4th generation

LSI-Large Scale integration, CP/M, MS DOS, GUI (macOS, Windows), Windows Server, Network operating systems, distributed operatings systems

History of Operating Systems (6)







Mobile Computers– 5th generation

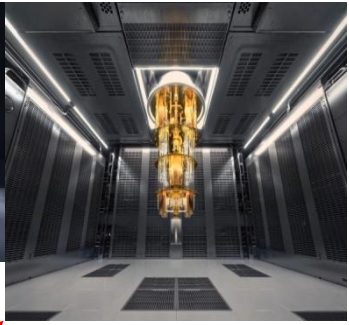
Symbian OS, RIM OS, Apple iOS, Android

Android , Linux based OS, is open source and available under permissive licence.

Hardware - Now and the Future?







The most advanced lithography machines, which are used to pattern millions of microcopic transistors, each smaller than a human cell, are made by ASML in the Netherlands. New NA EUV will produce for Intel next 2027 chips.



Cryogenics quantum computers, superconducting qubits need to be maintained at temperatures close to absolute zero (20-100 millikelvin) to minimize thermal noise and ensure quantum coherence. Even minute thermal vibrations at slightly higher temperatures, such as 4 Kelvin, can disrupt qubits, causing them to lose their quantum superposition states and revert to classical binary states through a process called decoherence.

First Full-Stack Silicon CMOS Quantum Computer - Quantum Motion